Allen Xu

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EDUCATION

University of California, San Diego **Sep. 2026 - Present**
Candidate for M.S. in Computer Science *San Diego, CA*

Northeastern University - *Khoury College of Computer Sciences* **Sep. 2022 - April 2026**
B.S. in Computer Science and Media Arts with Minor in Mathematics - Summa Cum Laude *Boston, MA*

- GPA: 3.97/4.00
- Dean's Scholarship: Merit scholarship awarded in recognition of outstanding academic achievement
- Courses: Multivariable Calculus, Discrete Differential Geometry, Object Oriented Design, Algorithms, Computer Graphics, Linear Algebra, Artificial Intelligence, Software Engineering, Probability and Statistics, Graph Theory

SKILLS

Languages: Java, Python, C++, TypeScript, Kotlin, SQL
Technologies: Docker, Kubernetes, AWS, PostgreSQL, React, PyTorch, Pandas, OpenGL
Software Development: REST APIs, Distributed Systems, Machine Learning, CI/CD, Git, Linux

PROFESSIONAL EXPERIENCES

Resmed | *Java, DataDog, Kubernetes, Postman* **May 2026 - Aug. 2026**
Software Engineer *San Diego, CA*

- Designed a context-object pattern to propagate data through legacy and modernized APIs, supporting 20M+ production API calls.
- Expanded Respiratory Care device support across multiple services, structuring feature flags to prep for FDA 510k approval process.
- Led a Kubernetes namespace migration by identifying the optimal cutover window, reducing the message backlog 94% (93K → 6K)
- Improved cross-team testing and engineering by building Playwright/Cucumber acceptance-test mocks adopted across 7 repositories and deploying DataDog latency monitors, reducing time-to-detect for service regressions.

Insulet | *Python, Docker, PostgreSQL* **June. 2025 - Aug. 2025**
Devops Engineer *San Diego, CA*

- Migrated live service Grafana database from SQLite3 to PostgreSQL, enabling scalability of **66+ tables** and an expanding user base.
- Containerized Grafana database services with Docker, improving modularity and deployment efficiency.
- Automated permission management by syncing LDAP's **2000+ employee** directory directly to Atlassian, reducing manual overhead

MORSE Corp. | *Python, Docker, Airflow* **Jan. 2025 - May 2025**
Software Engineer - Algorithmic Test and Evaluation *Boston, MA*

- Implemented lazy loading into the test and analysis pipeline, enabling support for **2-3 million rows** of data and improving transformation speed; focused on pipeline modularity to account for shifting analyses up to higher security levels for further analysis.
- Optimized **20+ step** analysis pipeline by utilizing aggregated data representations, reducing server overhead and improving efficiency.

Mirakl | *AWS, Python, Pandas, API* **Jan. 2024 - June 2024**
Software Engineer - Data, AI and Analytics *Boston, MA*

- Developed an XML sitemap parser to efficiently process **100K+ product entries**, replacing the company's reliance on Octoparse.
- Deployed cloud-based AWS applications continuously monitoring **50K+ product accounts**, enabling long term pattern detection.
- Built a pipeline to match **10K Walmart sellers** to Amazon, used OpenAI's o4 model and MistralAI's Large model for data parsing and relevance evaluation to aid in a **$1-2 million investmen**t decision by the Mirakl Labs team

Northeastern Khoury College of Computer Sciences | *Kotlin, Latex, Algorithms* **Sep. 2023 - Dec. 2024**
Teaching Assistant - Algorithms and Data Structures *Boston, MA*

- Provided personalized academic support through regularly-scheduled online one-on-one office hours to over **300 students**
- Planned and conducted weekly small-group lab sessions focused on reviewing challenging course concepts for groups of **30 students**

PROJECTS

Gaussian Splatting Streamliner | *Python, 3DGS* | Github Repo **Jan. 2026 - April. 2026**

- Packaged Reflct's sharp frame selection algorithm, RealityScan image alignment, and COLMAP-based exports into a single pipeline
- Pipeline resulted in ~3x reduction in image preprocessing time, enabling faster Gaussian Splatting workflows and iterations for users

Pac-man RL | *Python, PyTorch, Gymnasium, OpenAI ALE* | Github Repo **Oct. 2024 - Dec. 2024**

- Designed and implemented a Double-DQN with a Convolutional Neural Network (CNN) to train a Pac-Man agent
- Built a custom Gymnasium environment using Numpy to train ghost agents using a single-DQN with a Feedforward Neural Network (FNN), designed our network to predict the actions for 4 ghosts with a 16-dimensional output space representing movement options
- Integrated GPU acceleration with CUDA, hyperparameter tuning, and memory optimizations to **improve training speed by 2x**

Stack Underflow | *React, Node.js, OpenAPI, MongoDB, Socket.io, RL* | Website **Oct. 2024 - Dec. 2024**

- Integrated OpenAI to give suggestions during question creation and generate AI responses to questions to improve user engagement
- Designed and implemented features for creating or joining public and private communities, allowed users to share questions effectively.
- Ensured robust testing throughout development, maintaining a **95%+ overall test coverage** which greatly reduced errors